

Data Engineering Roadmap 2026

The following is the roadmap for a **Data Engineer**. It includes **FREE learning resources** for technical (tool) skills and soft (core) skills 🏃

This roadmap is based on the analysis of hundreds of data engineering jobs and our own experience in building production-grade data platforms and pipelines at AtliQ Technologies (<https://www.atliq.com>), where we have delivered over 25 data & AI projects in the last two years. More than 90% of our clients at AtliQ are **SMEs (Small to Medium-sized Enterprises)** based in the USA.

Link to data/data engineering jobs analysis:

https://github.com/codebasics/job-scrapper/blob/main/Analysis/Data%20Engineering/data_visualizer.ipynb

Data & AI case studies at AtliQ: <https://www.atliq.com/case-studies>

Link to our full YouTube content for this roadmap: [Link](#)

Find Your Suitability: Before starting your journey, it is important to understand whether this role truly aligns with your natural strengths and interests.

Take this short suitability test to identify if a Data Engineer career is the right fit for you:

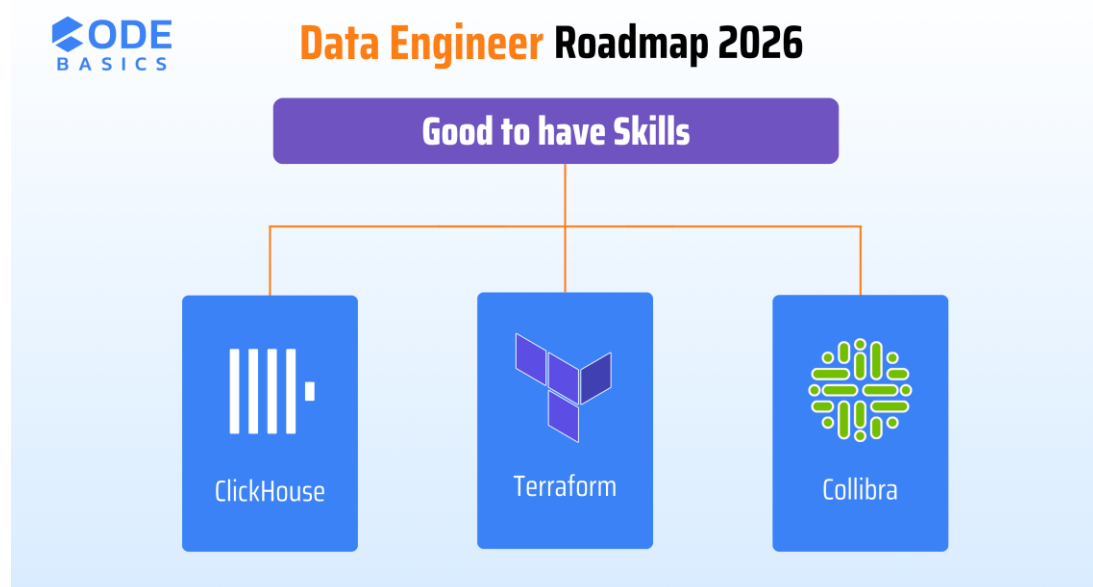
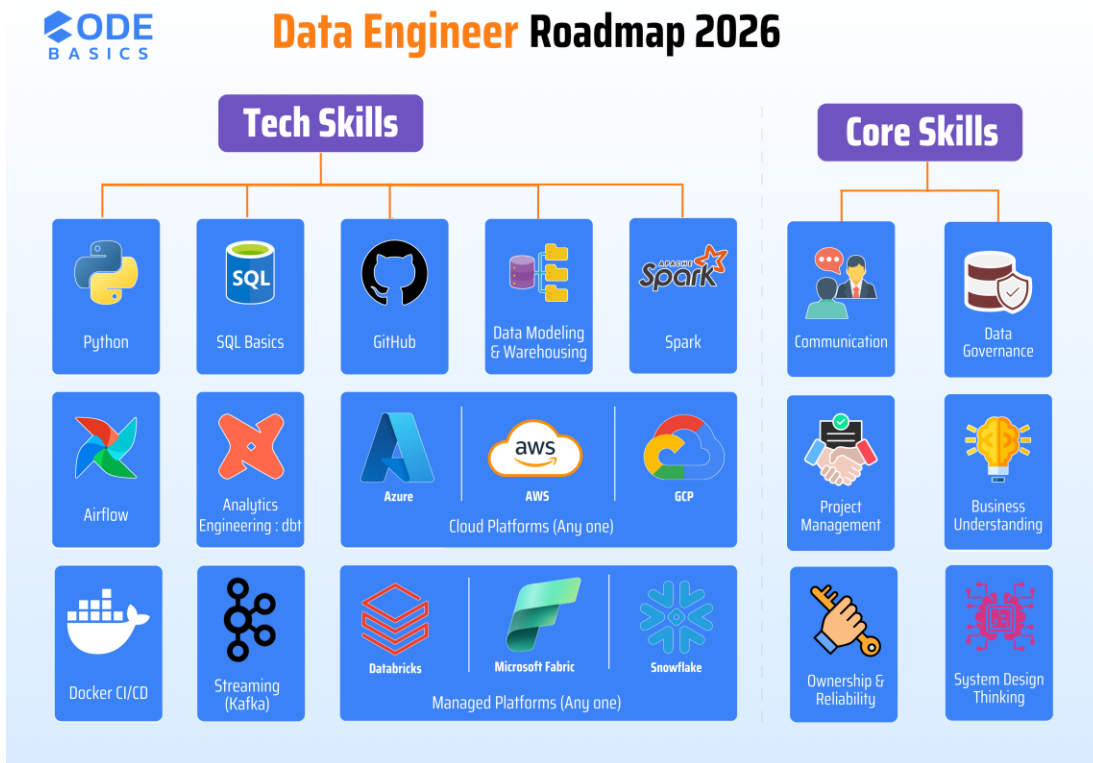
<https://codebasics.io/survey/find-your-match-de>

Proceed further if the results show that this career role matches you.

Total Duration: **6 to 8 Months** (4 hours of study every day, 6 days a week)

Roadmap is valid for **freshers** and **experienced professionals**.

Check the diagrams below for tech vs. core skills, as well as some good-to-have skills.



Week 0: Fundamental Research + Preparing your Mind & Body

Validate role fit, market demand, and prepare mentally for the journey

Research Questions Checklist:

- ☐ Is there growth in the market for this role?
- ☐ Are people getting jobs for this role?
- ☐ Is there reasonable pay for this role?
- ☐ Are there enough free resources for me to check my interest?
- ☐ Who are the big players (products) in my field of interest?
- ☐ Is there an active community of aspirants and professionals?
- ☐ What kind of problems are being solved in the industry using this field?
- ☐ Will this problem remain relevant in a few years to come with advancements in AI?
- ☐ Does this role align with my natural capabilities & interests?

WEF Report: https://reports.weforum.org/docs/WEF_Future_of_Jobs_Report_2025.pdf

Reality Check: https://www.linkedin.com/posts/hemvad_data-aspirants-know-this-reality-1-this-activity-7261582534826717184-9imE

Scam Awareness:

There are many active scams targeting aspirants at various stages of the journey

Scam Awareness Course: <https://codebasics.io/courses/scam-awareness-course>

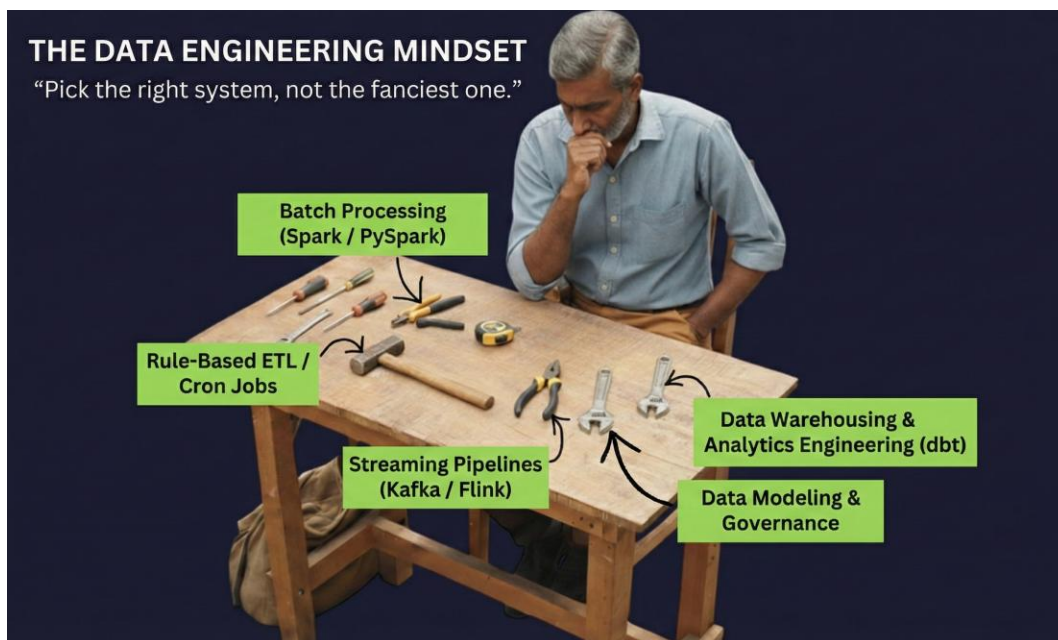
Preparing your Mind & Body

1. Accept that you are entering a competitive field
2. Train to use internet as your second brain
3. Disengage from “Victim” mindset people
4. Engage with “Hero” mindset people
5. Create a consistent sleep cycle
6. Eat mindfully, move regularly
7. Read any spiritual book

💡 **Insight:** Out of 100 aspirants, around 80 skip proper research. You're already among the 20 who start with clarity.

Mindset before we begin the learning journey ⌚

Have this mindset of an expert handyman, who has strong fundamentals, wide knowledge of tools and a judgement of picking the right tool at the right time.



Phase 1: Foundations

Week 1: Python Basics + Essential Data Structures

Learn Python programming basics for data engineering work.

Topics:

- Python basics (functions, loops, conditions)
- Lists, dicts, sets
- File handling: CSV, JSON
- Data structures: Arrays, Hash maps (very important), Stack, Queue, Big O notation

Learning Resources:

- Track A (Free)
 - Codebasics Python Tutorials (first 14 videos): [Link](#)
 - Corey Schafer Python tutorials (first 16 videos): [Link](#)
 - Data structures (First 8 videos) - [Link](#)
 - *Extremely Important:* Use ChatGPT, Claude.ai 🤖 as your personal tutor in case you have questions or facing issues
- Track B (Affordable Fees)
 - Python Course – [Link](#)

LinkedIn - Core Skill

- ☐ Create a professional-looking LinkedIn profile.
- ☐ Use this LinkedIn Checklist to update your profile: [Click here.](#)

Assignment:

- ☐ Track A: Finish all these exercises in the Python and Data structures tutorials playlist mentioned above in learning resources
- ☐ Track B: Finish exercises and quizzes for relevant topics

Why is LinkedIn important?

- Imagine you sell great samosas but open a shop in a quiet village, business struggles. You move to a busy city street, and suddenly business flourishes.

- LinkedIn is that busy street. A good profile is just opening the shop; real success comes from building strong online credibility (the tasty samosas), which we will discuss next.



- Avoid the mindset of “skills first, LinkedIn last.” Credibility takes time to build, so start working on it from Day 1 👍

Week 2 & 3: Python for Data Engineering

Python advanced programming for data engineering. You will do data transformations using Pandas this week, Pandas is for small data and prototyping. For large scale use cases, you will use Spark which we will cover later.

Topics:

- Exception handling and logging
- Virtual environments and pip
- OOP (classes, methods)
- Working with APIs using requests
- Dates, times, timezones
- Unit testing using pytest
- Pandas for data transformations

Learning Resources:

- Track A (Free)
 - Corey Schafer Python tutorials (Video 17, 20 to 26): [Link](#)
 - Codebasics Python Tutorials (Video 18, 35 to 38): [Link](#)
 - Pandas (Watch chapter 5 in this course, it is free): [Link](#)

- Track B (Affordable Fees)
 - Python Course – [Link](#)

Assignment:

□ Mini Project (OpsPulse): Pick a dataset (NYC Taxi, Public API – weather, finance). Fetch data from a public API or dataset. Save raw data to disk using JSON/CSV, transform it using Pandas. Add 2-3 basic pytest tests.

Week 4: Data Engineering Foundations + Git

Develop understanding of Data Engineering foundations and learn the Git version control system, which you will use to maintain and collaborate on your data engineering code.

Topics:

- Data Engineering Fundamentals
 - What is Data Engineering?
 - OLTP vs OLAP, data life cycle, SLA
 - ETL vs ELT
 - Batch vs Streaming
- Git
 - What is a version control system, Git and Github?
 - Git clone, add, commit, push, pull
 - Git branches, merge conflicts.

Learning Resources:

- Data Engineering
 - Data Warehouse vs Data Lake vs Data Lakehouse [Link](#)
- Git
 - Codebasics Git/GitHub Tutorial Playlist - [Link](#)
 - Corey Schafer YouTube tutorial - [Link](#)

Communication – Core skill:

- Start attending free Toastmasters' sessions to improve communication:
<https://www.toastmasters.org/>

Offline Community:

Here is an offline community made by data folks for the data folks

- Indian Data Club (Discord): <https://discord.com/invite/SQ3rU3ddFq>
- Indian Data Club (LinkedIn): <https://www.linkedin.com/company/indian-data-club>

Assignment:

- ☐ Move Week 1–2 code into a clean repo structure.
- ☐ Add README explaining:
 - What data is ingested
 - Why it exists, how it's transformed

Show your work:

- ☐ Share your GitHub repo on LinkedIn with:
 - What problem does it solve
 - One thing you learned about the DE fundamentals

Week 5 & 6: SQL Mastery

Become proficient in SQL for analytics and ETL

Topics:

- Relations database, tables, schemas (Use MySQL or PostgreSQL as a database)
- Basic queries: SELECT, WHERE, GROUP BY, ORDER BY, HAVING
- Joins (Inner, Left, Right, Outer)
- Aggregate functions (COUNT, SUM, AVG, MIN, MAX)
- Intro to DDL (CREATE TABLE, INSERT, UPDATE, DELETE)
- Window functions: ROW_NUMBER, RANK, LAG, LEAD, NTILE
- Common Table Expressions (CTEs)

Learning Resources:

- Track A (Free)
 - SQL Codebasics YouTube Tutorial - [Link](#)
 - SQL Khan Academy course - [Link](#)
 - Learn SQL in an interactive way - <https://sqlbolt.com/>
- Track B (Affordable fees)
 - SQL Codebasics Course - [Link](#)

Core/Soft Skills

- LinkedIn
 - Follow DE influencers
 - Zach Wilson: <https://www.linkedin.com/in/eczachly/>
 - Barra: <https://www.linkedin.com/in/baraa-khatib-salkini/>
 - Gowtham SB: <https://www.linkedin.com/in/sbgowtham/>
 - Increase Engagement
 - Start commenting meaningfully on Data Engineering and career-related posts.
 - Helps network with others working in the industry to build connections.
 - Learning and brainstorming opportunities.
 - Remember, *online presence is a new form of resume*
- Discord 
 - Start asking questions and get help from the community. This post shows how to ask questions the right way: <https://bit.ly/3I70EbI>
 - Join Codebasics Discord server: <https://discord.gg/r42Kbuk>

Business Understanding:

As a data engineer, you must understand how the business works and how data supports decision-making. This includes knowing business domains, key functions, how metrics are defined, and how analytics drives strategy. Strong business understanding helps you build pipelines that serve real business needs, not just move data.

Topics:

- Business domains and functions
- How data supports business decisions
- Translating business questions into data requirements
- Understanding metrics, KPIs, and definitions
- Analytics-driven competitive advantage
- Thinking beyond “data movement” to “business impact”

Learning Resources:

- How to build domain knowledge/Business Understanding: [Link](#)
- Get a basic understanding of Domains and Functions - [Link](#)
- Learn business concepts (Ex- ThinkSchool - [Link](#)):
 - How Hotstar Application Scaled 25 million Concurrent Users: [Link](#)

Assignment:

- ☐ OpsPulse project: Load processed data into a relational DB.

- ☐ Write 15–20 SQL queries covering joins, aggregations, filters, window functions, and CTEs
- ☐ Run one SQL query from Python and export CSV
- ☐ Write analytics queries using JOIN, GROUP BY, CASE, NULL handling, and window functions
- ☐ Solve a few Medium-level SQL problems on platforms like [Datalemur](#) / [HackerRank](#)
- ☐ Push SQL scripts and Python code to GitHub with a short README explaining the use case

Show Your Work:

- ☐ Write meaningful comments on at least 10 Data Engineering–related LinkedIn posts (e.g., pipelines, Snowflake, Databricks, Kafka, Airflow).
- ☐ Post:
 - One SQL problem you struggled with
 - Why your final query is safer than your first attempt
 (Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 so we can engage to increase your visibility)

Week 7 & 8: Data Modeling & Warehousing Basics + Linux Fundamentals

Learn data modeling and warehousing fundamentals. Learn Linux basics that will help you work with general data ops.

Topics:

- Data Modeling and Warehousing
 - Fact tables vs dimension tables
 - Star schema vs snowflake schema
 - SCD Type 1 and 2
 - Columnar storage, Partitioning, Compression
 - Layered architecture:
 - Raw / Staging / Core / Marts
 - Medallion architecture: Bronze / Silver / Gold
 - Basic data governance concepts:
 - Schema ownership
 - Naming conventions
 - PII (Personally Identifiable Information) awareness
- Linux
 - Basic Linux commands (cd, ls, mv, rm, chmod, grep, awk, sed)
 - Process management (ps, top, kill)
 - Shell scripting basics

Learning Resources:

- Track A (Free)
 - Star vs Snowflake Schema | Fact, Dimension Tables: [Link](#)
 - Medallion architecture: [Link](#) (Timestamp- 1:43:33 - 1:49:24)
 - Linux Fundamentals - [Link](#)
 - SQL Data Warehouse from Scratch | Full Hands-On DE Project - [Link](#)
- Track B (Affordable Fees)
 - Data Engineering Course – [Link](#)

Assignment:

- ☐ OpsPulse v2 Project: Design star schema (fact + dimensions)
- ☐ Map tables to DW layers (raw/core/mart).
- ☐ Define SCD logic for one dimension.

Show Your Work:

- ☐ Make a LinkedIn post explaining the assignment you have just completed.
(Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 if you want feedback/visibility)

💡 *Insight: Only **60 among 100** aspirants continue further. You be one among them 😊*

Phase 2: Cloud and Scale

Week 9 & 10: ETL on Cloud (Batch Pipelines)

Build and schedule a real batch ETL pipeline on your chosen cloud platform and learn how to make it reliable, repeatable, and cost-aware.

Topics:

- File formats: Parquet, Avro, JSON
- Data ingestion in Data Lake
- Batch ETL pipeline architecture
- Incremental vs full loads
- Idempotent batch processing
- Data partitioning strategy
- Logging and basic error handling
- Handling duplicate and late-arriving data
- Cost awareness in batch pipelines

Choose Your Cloud Platform (Pick One)

At this stage, pick one cloud platform and continue the rest of this week using the corresponding services.

The architecture and concepts remain the same, only the services change.

- Option A: AWS
 - Object storage: Amazon S3
 - Batch processing: AWS Glue
 - Ad-hoc queries using Athena
 - Amazon Redshift as a data warehouse
- Option B: Azure
 - Object storage: Azure Data Lake Storage (ADLS Gen2)
 - Azure Data Factory (ADF)
 - Synapse/Fabric
- Option C: GCP
 - Object storage: Google Cloud Storage (GCS)
 - Dataflow
 - BigQuery

Pick one track and implement the same batch ETL pipeline using the equivalent services.

Learning Resources:

- Azure for Data Engineers (YouTube): [Link](#)
- Azure for Data Engineer from Microsoft: [Link](#)
- AWS Data Engineer: [Link](#)
- Google cloud platform full course: [Link](#)

Ownership & Reliability

As a data engineer, you own pipelines that must run consistently without manual intervention. This includes handling failures, ensuring data correctness, managing re-runs, and being accountable for costs and reliability.

Topics:

- Ownership mindset in data engineering
- Pipeline failures and recovery
- Reliability vs speed trade-offs
- Idempotency and safe re-runs
- Cost and resource responsibility
- Thinking beyond “it works once”

Learning Resources:

- Pipeline failures and recovery - [Link](#)
- Reliability tradeoffs - [Link](#)
- Idempotency - [Link](#)
- Cost and resource responsibility - [Link](#)

Assignment:

- ☐ Pick one cloud platform (AWS / Azure / GCP)
- ☐ OpsPulse project: Move raw + processed data to cloud storage.
- ☐ Build a Python batch ETL job to ingest, transform, and load data
- ☐ Implement incremental loading
- ☐ Add logging and basic error handling

Show Your Work:

- ☐ Write a LinkedIn post using one of the following:
 - ☐ Identify 3 possible failure scenarios
 - ☐ Explain how your pipeline handles re-runs (idempotency)
- ☐ Document assumptions and failure handling in README

Week 11: Docker + ATS Resume

Build containerization skills for data engineering

Topics

- Images vs containers
- Dockerfile basics
- docker-compose
- Environment variables
- Skip: Docker internals, Kubernetes (for now)

Learning Resources

- Docker in one hour (YouTube): [Link](#)

Research:

- ☐ What are the projects your professional connections are working on?
- ☐ What are the 2 major reasons professionals are growing in their careers?
- ☐ What strategies do people who land jobs use?
- ☐ How do Data engineers collaborate with Data Analysts and Data scientists/AI engineers in real projects?

Presentation - Core Skills

- Learn presentation skills: Death by PowerPoint: [Link](#)

ATS Resume Preparation

- Resumes are dying but not dead yet. Focus more on online presence.
- Here is the resume template you can use it: <https://codebasics.io/resources/data-engineering-resume-template>
- Use this checklist to ensure you have the right ATS Resume: [Check here.](#)

Assignment:

- ☐ OpsPulse project: Containerize a Python ETL application and deploy with Docker Compose
- ☐ Update resume with this project

Week 12 & 13: Spark + Databricks

Use Spark + Databricks to process large datasets, understand distributed data processing deeply, and learn how to build scalable batch pipelines.

Topics:

- Spark & Distributed computing
- Databricks as a managed platform
- Spark dataframes
- Unity catalog and medallion architecture
- Transformations vs actions and lazy evaluation
- Narrow vs wide transformations
- Joins and aggregations at scale
- Partitioning and parallelism
- Shuffles and performance implications
- Caching and persistence
- Incremental batch processing with Spark

Learning resources:

- Databricks free crash course on YouTube: [Link](#)

System Design Thinking

When data grows, single-machine solutions stop working. Spark forces you to think in terms of distributed systems, data movement, and trade-offs between simplicity, performance, and scalability.

Topics:

- Single-node vs distributed systems
- Horizontal scaling concepts
- Data movement and bottlenecks
- Trade-offs between performance and complexity
- Choosing Spark vs simpler solutions

Learning resources:

- System Design YT Playlist: [Link](#)
- Engineering blogs on large-scale data processing (Databricks) - [Link](#)
- Talks on distributed systems and data pipelines - [Link1](#), [Link2](#)
- System design discussions focused on data platforms - [Link](#)

Assignment:

- ☐ Set up Spark using [Databricks Free Edition](#)

- ☐ Load a large dataset using Spark DataFrames (for OpsPulse project)
- ☐ Apply transformations, joins, and aggregations
- ☐ Write output data in partitioned format
- ☐ Experiment with caching or repartitioning

Show your work:

- ☐ Explain why Spark is required for this case on a LinkedIn post.
- ☐ Identify one performance bottleneck Spark helps solve on a LinkedIn post.
- ☐ Document design trade-offs in README
(Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 so we can engage to increase your visibility)

Week 14 & 15: End-to-End Data Engineering Projects on Databricks

Complete two end-to-end projects (preferably in different domains – FMCG, Retail, Healthcare, Transportation, etc.) using Databricks. The project should cover the following topics,

Topics:

- Data ingestion from OLTP to OLAP
- Medallion processing through bronze, silver and gold layers
- Spark Declarative Pipelines (SDP)
- Full vs incremental load
- Schema evolution
- Backward/forward compatibility

Learning resources:

- End-to-end data engineering project in the transportation domain: [Link](#)
- Spark Declarative Pipelines – Databricks End-to-End Project: [Link](#)
- IPL Data Analysis | Apache Spark End-To-End Project: [Link](#)

Assignment

- ☐ Complete the assignments given in those YouTube projects.

Show your work:

- ☐ Write LinkedIn posts sharing your learning on those projects
(Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 if you want feedback/visibility)

Week 16 & 17: Snowflake + Airflow

Build production-style data pipelines using a managed data platform and orchestration, while learning how to plan and deliver data projects effectively.

Topics:

- Snowflake
 - Cloud data warehouse basics with Snowflake
 - Staging → Core → Analytics layers
 - Zero-copy cloning
 - Incremental patterns (streams/tasks/dynamic tables conceptually)
 - Data loading patterns (COPY INTO, file format configs)
- Airflow
 - Airflow architecture
 - DAG, Task, Operator
 - Hook, Provider, Task Group
 - XComs for task communication

Learning resources:

- Snowflake data engineer academy tutorial on YT: [Link](#)
- Airflow for beginners: [Link](#)

Project Management

As pipelines grow, work needs to be planned, sequenced, and delivered reliably. Project management helps you break work into tasks, manage dependencies, and deliver data solutions on time.

Topics:

- Breaking data projects into tasks
- Managing dependencies between steps
- Estimating effort and timelines
- Tracking progress
- Handling scope changes
- Coordinating work across systems

Learning resources:

- Scrum: <https://scrumtrainingseries.com/>
- Kanban: <https://youtu.be/jf0tlbt9lx0>
- Tools: JIRA, Notion

Motivation:

Data Engineer At Grubhub After 6 Years Of Career Break: [Link](#)

Assignment:

- ☐ Pick one managed platform (Snowflake / Databricks / Fabric)
- ☐ Load data from cloud storage into the platform
- ☐ Apply transformations inside the platform
- ☐ Orchestrate the pipeline using Airflow
- ☐ Schedule and monitor the pipeline

Show your work:

- ☐ Push code and configs to GitHub
- ☐ Break the pipeline into clear tasks
- ☐ Document dependencies between tasks
- ☐ Define a simple execution plan in README
- ☐ Create a LinkedIn post by sharing your GitHub Repo.

(Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 if you want feedback/visibility)

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Phase 3: Advanced & Specialization

Week 18 & 19: Streaming Data Systems: Kafka + Flink

Build a basic real-time data pipeline using Kafka and learn when streaming is the right choice over batch processing.

Topics:

- Streaming fundamentals
 - Batch vs streaming processing
 - Real-time data use cases
 - Event-driven architecture
- Kafka
 - Kafka architecture (broker, topic, partition)
 - Confluent as a managed Kafka platform
 - Producer, consumers
 - Message keys and partitions
 - Ordering and parallelism (conceptual)
- Flink
 - Flink architecture
 - Watermarks, checkpoint, fault tolerance
 - Flink SQL
- Streaming Pipelines
 - Build a simple producer and consumer
 - Process streaming data
 - Handle duplicates and late events
 - Reprocess events (conceptual)

Learning resources:

- Kafka crash course: [Link](#)
- Flink tutorial playlist: [Link](#)

System Design Thinking

Streaming systems introduce new trade-offs around latency, consistency, and complexity. You need to reason about when real-time processing is required and when batch is the better choice.

Topics:

- Batch vs stream processing
 - Latency vs complexity

- Exactly once vs at-least-once (conceptual)
- Designing for failure in real-time systems
- Choosing streaming only when necessary

Learning resources:

- Batch vs streaming process - [Link](#)
- Latency vs complexity - [Link](#)
- Exactly once vs at-least-once (conceptual) - [Link](#)

Assignment:

- ☐ Set up Kafka locally or using a managed service such as confluent
- ☐ Create a producer to publish events
- ☐ Create a consumer to process events
- ☐ Process streaming data and write output to storage

Show your work:

- ☐ Explain why streaming is required for this case on a LinkedIn Post
- ☐ Compare this pipeline with a batch alternative on a LinkedIn Post
- ☐ Document design trade-offs in README

Weeks 20 & 21: Analytics Engineering with dbt + Cloud Scaling + Portfolio

Transform raw data into analytics-ready models using dbt and learn how to scale and govern analytics workloads on the cloud.

Topics:

- Analytics Engineering
 - Role of analytics engineering in the data stack
 - Transforming raw data into analytics-ready models
 - dbt fundamentals:
 - Models
 - Sources
 - Tests
 - Incremental models
 - Building analytics layers on top of warehouses
- Cloud scaling
 - Running analytics workloads at scale
 - Separating storage and compute
 - Managing environments (dev / prod – conceptual)
 - Cost awareness for analytics workloads

- Scaling pipelines as data grows

Learning resources:

- dbt official tutorials: [Link](#)
- dbt fundamentals: [Link](#)
- Netflix Data Analysis | End-To-End Project: [Link](#)

Portfolio Building Resources:

You need a portfolio website in 2026. You can build your portfolio by using these free resources.

[Datascience portfolio](#): Helpful in creating a quick static Portfolio

[Linktree](#): Helpful to add multiple links on one page.

[GitHub](#): Another way to add projects and showcase it to hiring managers.

Codebasics Portfolio Website:

This is part of Codebasics Data Engineer Bootcamp.

Example Portfolios: <https://codebasics.io/portfolio/>

Data Governance

As data becomes business-critical, you need clear ownership, definitions, and controls to ensure consistency, trust, and compliance across teams.

Topics:

- Data ownership and responsibility
- Defining and documenting metrics
- Schema consistency and evolution
- Basic governance practices in analytics
- Preventing metric and data misuse

Learning resources:

- Data Governance Explained by IBM: [Link](#)
- Blogs and talks on data governance and analytics best practices - [Link](#)

Assignment:

- ☐ Set up dbt on top of your chosen warehouse
- ☐ Create analytics models from raw or processed data
- ☐ Implement tests for key models
- ☐ Generate dbt documentation
- ☐ Run analytics workloads and observe costs
- ☐ Define ownership for key datasets

Show your work:

- ☐ Make a LinkedIn post on the following items:
 - ☐ Document metric definitions and assumptions
 - ☐ Explain how governance improves trust in your pipeline
- ☐ Add a link to your projects in your resume and LinkedIn.

(Tag [Codebasics](#), [Dhaval Patel](#) and [Hemanand Vadivel](#) with the hashtag #deroadmap26 if you want feedback/visibility)

Weeks 22 & 23: Data Quality & Reliability

Topics:

- Importance of data quality in production systems
- Types of data quality checks:
 - Schema checks
 - Null checks
 - Uniqueness checks
 - Volume checks
- Implementing data quality checks in analytics pipelines
- Monitoring data freshness
- Handling data quality failures
- Re-running pipelines safely after failures
- Observability basics for data pipelines (conceptual)

Learning resources:

- Data Quality Explained by IBM - [Link](#)
- How to solve data quality issues | Data Reliability Fundamentals Book Author: [Link](#)
 - Data Quality Fundamentals Book: [Link](#)
- dbt tests - [Link](#)
- dbt documentation - [Link](#)
- Blogs on data quality and observability in data engineering - [Link](#)

Ownership & Reliability

At this stage, your responsibility is not just to build pipelines, but to ensure the data they produce is trustworthy, consistent, and reliable over time.

Topics:

- Taking ownership of data correctness

- Proactively identifying data issues
 - Responding to data incidents
 - Building trust with data consumers

Learning resources:

- Engineering blogs on data incidents and reliability - [Link](#)

Motivation:

Data Engineering Career Tips By Airbnb Data Engineer: [Link](#)

Assignment:

- ☐ Add data quality checks to an existing pipeline
- ☐ Implement schema and null checks
- ☐ Track data freshness for key datasets
- ☐ Identify common failure scenarios in your pipeline

Share your work:

- ☐ Document how issues are detected and handled
- ☐ Explain how these checks improve trust in your data in a LinkedIn post

💡 *Insight: Only **30 among 100** aspirants continue further. You be one among them 😊*

Phase 4: Portfolio & Job Hunt

Weeks 24-26: **Business Projects, CI/CD & Open Source**

Build end-to-end data engineering projects, automate them with CI/CD, and make your first meaningful open-source contribution.

Topics:

- Designing an end-to-end data engineering project
- Integrating batch, streaming, and analytics pipelines
- Using GitHub for collaboration and version control
- CI/CD basics for data pipelines
- Automating pipeline execution and validation
- Reading and understanding large codebases
- Contributing to open-source data engineering projects

Learning resources:

- Git/GitHub full playlist - [Link](#)
- Airbnb Project | DBT + Snowflake + AWS: [Link](#)
- DE Projects on Codebasics YT Channel: [Link](#)
- More DE Projects: [Link](#)

At this stage, you're capable of building projects without step-by-step guidance.

You may use guided projects for practice, but to truly stand out, define and build at least one project on your own:

- Identify a business problem
- Source the data (API, public dataset, logs, etc.)
- Design and build the pipeline end to end

Most candidates submit identical guided projects.

Self-defined projects signal real engineering thinking and ownership.

💡 Insight: Guided projects teach tools. Self-defined projects prove you're job-ready.

Where to Find Self-Defined Project Ideas

If you're unsure how to define your own project, use these high-signal sources:

- Your own pain points

- Repetitive reports, manual CSV handling, delayed dashboards, unreliable data.
- Public APIs
 - Build pipelines using real data (finance, weather, mobility, sports, logs).
- Company tech blogs & case studies
 - Recreate a simplified version of how a company solved a data problem.
- Your guided projects extend them:
 - add a new data source
 - change the business question
 - add reliability, quality, or scale constraints
- Use your AI friend (ChatGPT, Claude, Gemini) to generate and refine project ideas

Why these (open source contribution) matters

This is exactly how engineers move into open source:

- Start with a personal or business use case
- Build a solution
- then, contribute improvements back to existing projects

💡 Insight: Many open-source contributions begin as self-defined projects.

Start contributing to opensource

- Start contributing to open source (Data Engineering)
 - Check this video to get started: [Link](#)
- Use ChatGPT as your friend to understand the codebase, issues, and how to raise PRs.
- Find any data engineering–related repository (Airflow, dbt, Great Expectations, etc.), check Issues and filter for “good first issue”.
- You can also try some simpler / learning-focused repositories such as:
 - <https://github.com/DataTalksClub/data-engineering-zoomcamp>
 - <https://github.com/astromer/astro-sdk>
 - https://github.com/great-expectations/great_expectations

Assignment:

- ☐ Build and document at least one unguided end-to-end data engineering project. Extend OpsPulse with one new requirement: new source or scale constraint or reliability feature
- ☐ Integrate multiple tools from the roadmap
- ☐ Set up basic CI/CD
- ☐ Push complete project to GitHub
- ☐ Pick one data engineering open-source project
- ☐ Understand the project structure

- ☐ Identify a small contribution: Documentation fix, or Minor bug fix, or Small feature enhancement
- ☐ Raise at least one pull request
- ☐ Break work into milestones
- ☐ Track progress and incorporate feedback
- ☐ Document decisions and learnings

Share your work:

- ☐ Make a LinkedIn project post with a video presentation explaining the one project as if you are explaining it to the stakeholders. Example post: <https://bit.ly/47WbRG9>

Weeks 27-30: Personal Branding, Interviews & Long-Term Career Growth

Prepare for data engineering interviews using the STAR method, practice advanced SQL interview questions, and build a personal brand and communication skills that support long-term career growth.

Topics:

Interview Preparation – Data Engineering

- SQL interview patterns:
 - Complex joins
 - Window functions
 - Subqueries and CTEs
 - Edge cases and NULL handling
- Python interview patterns:
 - Data processing logic
 - Error handling
 - Reasoning through edge cases
- Data engineering system design interviews:
 - Batch pipeline design
 - Streaming vs batch decisions
 - Storage and compute choices
 - Failure handling and backfills
 - Explaining projects clearly
 - Discussing trade-offs and decisions

Learning resources:

- Data engineering interview blogs and GitHub repositories

- YouTube mock interviews for data engineers
- System design discussions focused on data pipelines
- SQL Interview Practice Platforms:
 - HackerRank (SQL – Advanced)
 - LeetCode (SQL – Medium to Hard)
 - DataLemur (SQL Interview Questions)

Communication & Personal Branding

To grow long-term in your career, you need to clearly communicate your experience, decisions, and impact. Interviews and visibility both depend on how well you structure and present your story.

Topics:

Interview Communication:

- STAR framework:
 - Situation
 - Task
 - Action
 - Result
- Structuring technical and behavioral answers using STAR
- Explaining failures, trade-offs, and learnings
- Communicating ownership and impact

Personal Branding:

- Positioning yourself as a data engineer
- Showcasing projects with outcomes, not just tools
- Sharing learnings publicly with consistency
- Communicating with Stakeholders

Learning Resources

- Track A (Free)
 - STAR Method: [Link](#)
 - Tips for Designing Questions for Stakeholder Conversations: [Link](#)
 - What are Stakeholders?: [Link](#)
 - Check this project video on how to communicate with a Stakeholder: [Link](#)
- Track B (Affordable Fees)
 - Mastering Communication & Stakeholder Management Course: [Link](#)
 - Personal Branding for Professional Course: [Link](#)

Job Applications:

- Cold emailing: https://youtu.be/tD_sQSzq6uY

- Job referrals. Use LinkedIn as much as you can!

Follow this method to approach professionals: <https://bit.ly/3tezRFi>

Assignment:

- ☐ Solve 30–40 SQL interview questions across:
 - HackerRank
 - LeetCode
 - DataLemur
- ☐ Focus on medium to hard difficulty
- ☐ Write short explanations for your SQL approach and edge cases as a LinkedIn Post
- ☐ Prepare 5–7 interview stories using the STAR method:
 - One end-to-end pipeline
 - One failure or outage
 - One trade-off decision
- ☐ Practice explaining one project end-to-end using STAR
- ☐ Pick any domain (fintech, banking, healthcare, etc.) and write **5–7 clear questions** you would ask a stakeholder to understand their requirement.
- ☐ Post your work at least once in 2 weeks on LinkedIn.

Example post: <https://bit.ly/3WMTgGK>
- ☐ Make at least two comments per week.
- ☐ Discord server participation

💡 *Insight: Only 10 among 100 aspirants continue further. You be one among them 😊*

Supplementary Learning:

If you have additional time, consider learning these supplementary learning to stand out in the job market. They not only enhance your current skill set but also open up new career paths.

“Certifications are optional and should be attempted only after building hands-on projects from the roadmap”

DP 600 / 700 Certification:

Hiring managers often use certifications to filter candidates. Nowadays, obtaining a certification online is easy, so it's important to choose ones that are more challenging to earn. Two such certifications are,

DP-600 Microsoft Certification for Fabric Analytics Engineer Associate: [Link](#)

DP-700 Microsoft Certification for Fabric Data Engineer Associate: [Link](#)

These certifications are not easily attained and require a solid level of Microsoft Fabric skills, meeting the criteria that many hiring managers look for.

Cloud fundamentals certifications:

Cloud fundamentals certifications signal basic cloud literacy and familiarity with how data systems run on modern cloud platforms. They help hiring managers assess readiness for cloud-based roles, especially for candidates from non-cloud backgrounds, and add credibility when paired with hands-on projects.

Pick any one of the following:

- AWS Certified Cloud Practitioner: [Link](#)
- Azure Fundamentals (AZ-900): [Link](#)
- Google Cloud Digital Leader: [Link](#)

dbt Fundamentals Certification Course: [Free]

Reinforces analytics engineering concepts such as data modeling, testing, and documentation, and signals your ability to build analytics-ready data layers used by business teams.

[Course & Certification Link](#)

Databricks Official Certifications:

FREE Certifications:

There are many Free certifications available on the Databricks platform on specific topics. You can consider finishing the course and appearing for the certification.

Explore more here - [Link](#)

Databricks Certified Data Engineer Associate:

The Databricks Certified Data Engineer Associate certification exam assesses an individual's ability to use the Databricks Data Intelligence Platform to complete introductory data engineering tasks.

Explore more here - [Link](#)

Fundamentals of Data Engineering Book - Joe Reis & Matt Housley

Practical perspective on modern data engineering roles, responsibilities, and system design.

[Book Link](#)

Designing Data-Intensive Applications Book - Martin Kleppmann

Builds strong intuition around data systems, trade-offs, and architecture. High signal for long-term growth.

[Book Link](#)

How can I access guided learning all in one place with practical job assistance?

If you need guided learning with high-quality learning resources along with practical job assistance, then check this affordable Data Engineering Bootcamp by Codebasics:
<https://codebasics.io/bootcamps/data-engineering-bootcamp-with-virtual-internship>

The bootcamp will help you save time compared to free learning resources. Also, it will provide job and interview assistance, along with 2 virtual internships 🔥👩💻

Tips for effective learning

Spend less time-consuming information, more time in

- Digesting
- Implementing
- Sharing

Group Learning

Use **partner-and-group-finder** channel on Codebasics Discord server for group study and hold each other accountable for the progress of your study plan. Here is the Discord server link:
[Discord](#)

Final Checklist/ Concluding Actions Checklist:

The more boxes you tick, the closer you get to opening the door to your first Data Engineer job!



DATA ENGINEER ASPIRANT CHECKLIST 2026



SQL	PYTHON	DE FOUNDATIONS
☐ COMPLEX JOINS & WINDOWS ☐ ANALYTICAL QUERY WRITING ☐ DATA VALIDATION CHECKS ☐ QUERY PERFORMANCE BASICS	☐ MODULAR DATA PROCESSING ☐ FILES & DATA FORMATS ☐ PANDAS / PYSPARK USAGE ☐ LOGGING & ERROR HANDLING	☐ END-TO-END PIPELINES ☐ BATCH VS STREAMING ☐ ETL VS ELT ☐ DATA QUALITY BASICS
GIT	SPARK	SNOWFLAKE
☐ VERSION CONTROL WORKFLOWS ☐ BRANCHING & PULL REQUESTS ☐ CODE REVIEWS ☐ SAFE ROLLBACKS	☐ DISTRIBUTED PROCESSING BASICS ☐ TRANSFORMATIONS VS ACTIONS ☐ JOINS & SHUFFLES ☐ PERFORMANCE TUNING	☐ CLOUD WAREHOUSE CONCEPTS ☐ DATA LOADING & TRANSFORMS ☐ COST-AWARE QUERYING ☐ WAREHOUSES & SCHEMAS
AIRFLOW	DBT	CLOUD PLATFORM (AWS / AZURE / GCP)
☐ DAG DESIGN ☐ SCHEDULING & RETRIES ☐ BACKFILLS & RERUNS ☐ PIPELINE MONITORING	☐ SQL TRANSFORMATIONS ☐ INCREMENTAL MODELS ☐ DATA QUALITY TESTS ☐ DOCUMENTATION & LINEAGE	☐ OBJECT STORAGE ☐ COMPUTE VS STORAGE ☐ IAM BASICS ☐ COST OPTIMIZATION
DOCKER & CI/CD	KAFKA & FLINK	LINUX & DSA
☐ CONTAINERIZED PIPELINES ☐ ENVIRONMENT CONSISTENCY ☐ CI/CD WORKFLOWS ☐ AUTOMATED DEPLOYMENTS	☐ EVENT-DRIVEN SYSTEMS ☐ KAFKA PRODUCERS/CONSUMERS ☐ STREAM PROCESSING ☐ DELIVERY GUARANTEES	☐ LINUX COMMAND BASICS ☐ FILE & PROCESS HANDLING ☐ DATA STRUCTURE FUNDAMENTALS ☐ ALGORITHMIC THINKING BASICS

DATA ENGINEER ASPIRANT CHECKLIST 2026



SYSTEM DESIGN THINKING	OWNERSHIP & READABILITY	BUSINESS UNDERSTANDING
☐ END-TO-END DATA DESIGN ☐ BATCH VS STREAMING ☐ SCALABILITY & RELIABILITY ☐ FAILURE RECOVERY STRATEGIES	☐ PIPELINE OWNERSHIP MINDSET ☐ CLEAN, READABLE CODE ☐ NAMING & STRUCTURE ☐ MAINTAINABLE DATA WORKFLOWS	☐ BUSINESS METRICS AWARENESS ☐ DATA IMPACT UNDERSTANDING ☐ COST VS VALUE THINKING ☐ TRANSLATING BUSINESS NEEDS
PROJECT MANAGEMENT	DATA GOVERNANCE	COMMUNICATION
☐ TASK BREAKDOWN & PLANNING ☐ DEPENDENCY MANAGEMENT ☐ DELIVERY TIMELINES ☐ RISK & ISSUE TRACKING	☐ DATA QUALITY STANDARDS ☐ SCHEMA & CONTRACT MANAGEMENT ☐ ACCESS CONTROL BASICS ☐ COMPLIANCE & AUDITS	☐ EXPLAINING DATA PIPELINES ☐ STAKEHOLDER UPDATES ☐ TECHNICAL DOCUMENTATION ☐ CROSS-TEAM COLLABORATION
RESUME	LINKEDIN	PORTFOLIO
☐ PROJECT-FOCUSED RESUME ☐ STAR METHOD ☐ IMPACT-DRIVEN BULLETS ☐ ATS-FRIENDLY FORMATTING	☐ GOOD DP + BANNER ☐ 10 ENDORSED SKILLS ☐ 3 RECOMMENDATIONS ☐ 2 PROOF-OF-WORK POSTS	☐ END-TO-END DATA PROJECTS ☐ PRODUCTION-STYLE PIPELINES ☐ CLOUD & BIG DATA ☐ GITHUB WITH DOCUMENTATION

All the best! 🍑 🍑

